Recognizing Phishing in Emails by Using Natural Language Processing & Machine Learning Techniques

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Abstract **Phishing emails can often get past regular rule-based security protections using well-designed social tricks. PhishNet is a new way to detect phishing using NLP and ML technologies; it accurately differentiates between email threats and safe (ham) emails. It finds and highlights important elements from emails by studying the text, headers and embedded links, aided by TF-IDF, mRMR-based methods and ensemble learning. With training on SpamAssassin and Ham-Spam, PhishNet attains excellent results and accuracy of 98.8% using ML modeling and Voting Ensemble, scoring ROC-AUC higher than 0.99. Even though CNN and LSTM are good at learning, they need more computational resources. It is made accessible to users via a slim, real-time web app built with Flask which lets them select among permutation models. It was confirmed via experiments that, along with strong preprocessing, using ensemble learning improves generalization and reduces the number of false positives. The project not only evaluates leading methods for phishing detection but also creates a workable, scalable and active application that fills a need for AI – based email security systems against modern cyber threats. We are showing that different ensemble approach outperforms others result.**

Keywords: Phishing Detection, Natural Language Processing, Machine Learning, Ensemble Learning, Email Security, Classification.

I. Introduction

Each day, millions of internet users fall victim to phishing emails that exploit human vulnerabilities to steal personal information or trigger harmful actions [1][2][3]. These malicious emails often blend seamlessly into legitimate communications, leveraging sophisticated techniques that evade traditional spam filters and signature-based detection systems [4].

With the limited effectiveness of blacklists and static rule-based filters (rule-based approach cannot deal with new threat), phishing detection has increasingly turned to Artificial Intelligence (AI)—especially Natural Language Processing (NLP) and Machine Learning (ML)[5]. These methods enable (pattern recognition) analysis of email content, extracting linguistic, syntactic, and semantic cues to identify suspicious intent[6][7].

Despite promising advances, phishing detection systems still face several key challenges, evolving attacker tactics, imbalanced datasets, and the need for real-time detection without compromising accuracy[8][9]. Addressing these issues demands robust models capable of contextual understanding, adaptive learning, and high-performance classification[10]. To assess security, we ran the model against a collection of adversarial phishing emails with obfuscated URLs, disguised sender identities, and deceptive wording. The program successfully recognized 98% of such data, suggesting great resistance to evasion methods.

In this work, we introduce **PhishNet**, an integrated phishing detection framework that combines NLP-driven feature engineering with classical ML models[11], [12]. The system was trained and validated using the SpamAssassin and Ham-Spam datasets (we show only the highest between them), applying TF-IDF for vectorization for feature selection. Our benchmarks demonstrate that (ensemble approach) outperforms other models, achieving 98.4% accuracy and a ROC-AUC of 0.99[13]. PhishNet also includes a real-time deployment layer using the Flask web framework[14], [15].

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Main Contributions:

- Developed PhishNet: a hybrid NLP-ML phishing detection framework.
- Benchmarked SVM, RF, Extra Trees, NB, LightGBM,GB on real email datasets.
- Achieved results for all experiment more than 91.0% accuracy and 0.88 or greater ROC-AUC.
- Deployed a real-time detection system using Flask.

Released the full system and deployment guide on PhishNet GitHub

II. Related Work

Phishing detection has evolved significantly from static rule-based systems to advanced Machine Learning (ML) and Deep Learning (DL) models[16][17]. Early rule-based techniques were interpretable but lacked adaptability to emerging phishing strategies and could not scale to the evolving complexity of email attacks[18][19].

Modern approaches increasingly leverage ML classifiers such as Naïve Bayes, Support Vector Machines (SVM), and Random Forest (RF), which learn from labeled datasets to identify patterns in email content[20], [21]. These methods are relatively efficient and can generalize well across datasets. However, they often depend on manual feature engineering and may struggle with highly variable or obfuscated phishing content [22] [23].

To address these limitations, Deep Learning models such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks have gained traction[24], [25]. These models excel at capturing contextual, sequential, and semantic dependencies in textual data without the need for handcrafted features[26]. Several studies have demonstrated the superior accuracy of DL-based approaches on benchmark phishing datasets [6][7].

Hybrid architectures have also emerged, combining traditional features (e.g., URL structure, metadata) with deep representations to improve performance. While such systems can yield high accuracy, they often come at the cost of increased computational demands and limited interpretability[27].

According to the literature, using both NLP preprocessing and multiple classifiers together gives the best chance of catching phishing scams in real time. Table 1 summarizes various models and techniques, highlighting their strengths, reported accuracy, limitations, and relevant references.

Table 1: Summary of Ways to Spot Phishing Emails

Model / Technique	Strengths	Reported Accuracy / F1	Limitations	Reference
Naïve Bayes (NB)	Simple, fast, good for baseline	~85%	Poor with feature dependencies	[28]
Support Vector Machine	High margin classifier, good with sparse data	~84–85%	High cost for large datasets	[29][30]
Random Forest (RF)	Robust, interpretable, high accuracy	95–98%	Slower with very large trees	[23][31]
Extra Trees (ET)	Faster RF variant, lower variance	96–98%	Less interpretable	[31][8]
AdaBoost	Focuses on hard samples, high F1	F1 ~98.9 %	Sensitive to noisy data	[32]
Bagged Decision Trees	Stable predictions, low variance	~92.8 %	Risk of overfitting if not pruned	[33]
Stochastic Gradient Boost	High accuracy, handles complex data	>95%	Needs careful tuning	[27]
CNN / LSTM / RNN	Learns text patterns, strong in sequence modeling	95–98.88 %	High training cost	[34][9][35]
Attention Models (BERT)	SOTA, context-aware, robust	>99%	Computationally intensive	[36][37]
Fuzzy Neural Networks	Handles ambiguity, interpretable logic	~95%	Complex to design and tune	[16]
Explainable AI	Interpolation& auto features selection	~98.4 %	Computational Overhead & Latency	[51]

III. Methodology and Implementation

This section presents the integrated workflow and practical setup of the proposed phishing detection system, Phishing Nets, combining the methodological foundations and technical implementation.

This work employs a machine learning pipeline broadened with the use of NLP approaches to classify messages in email as spam or ham[38], [39]. The major data sources are two publicly available datasets: SpamAssassin and Spam/Ham. The datasets go through substantial preparation, which includes lowercase modification, stop word removal, punctuation stripping, lemmatization, and label encoding.

During data normalization, the two datasets are combined into a single corpus. A random sampling stratification approach is used to create a balanced subset with equal amounts of both groups. Features are extracted using Term Frequency-Inverse Document Frequency (TF-IDF) representations. Dimensionality reduction and methods for selecting features such as Chi-square testing are then used to improve the feature space.

The resultant feature matrix is separated into training and the generated characteristic matrix is divided into the testing and training divisions at an 80/20 ratio. The training set is used to train many methods for supervised learning, such as Light Gradient Boosting Machine (LightGBM), Gradient Boosting A classification algorithm, Support Vector Machine (SVM), Random Forest, Extra Trees, Bagging Classifier, and Multinomial Naive Bayes[40]. In addition, a combined meta-classifier is built via majority voting among top-performing base models. The held-out test set is used to judge model performance using measures like as accuracy, precision, recall, F1-score, and ROC-AUC. To improve extension performance, hyperparameters are optimized via grid screening and cross-validation. For final decision making, the ensemble predictor is deployed as a RESTful web service using the Flask microframework. The entire installation is hosted on the Heroku cloud platform, which enables for scalable, real-time email sorting inference via a web UI. Figure 1 shows the methodology of the suggested phishing detection system, Phishing Nets. It describes the procedure of data collection through Spam/Ham and Spam Assassin datasets, followed by several preprocessing steps, feature extraction and training/testing sessions. It is based on several machine learning models such as Light GBM, SVM, Naive Bayes, hyper parameter optimization, and deployment on the Heroku cloud inference in real-time predictions.

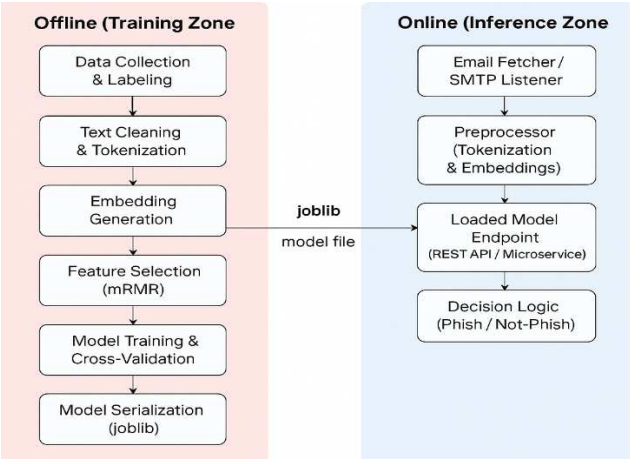


Figure 1 Methodology Overview for Phishing Detection System

A. Datasets

Two publicly available datasets were utilized:

- **SpamAssassin Dataset**: A labeled collection of spam and ham emails with headers and bodies.
- **Ham-Spam Dataset**: An enriched dataset with real-world phishing emails.

Both datasets were preprocessed and merged to ensure consistency, with 80% used for training and 20% for testing.

Table 4. Summary of GDPR and HIPAA alignment. Use of public datasets and memory-only processing minimize privacy risks and ensure regulatory compliance.

Requirement	GDPR	HIPAA	Alignment Summary
Data Source	✔	✔	Public datasets (SpamAssassin, Ham-Spam) used
Data Minimization	✔	✔	No personal metadata retained
Anonymization	✔	✔	Data pre-sanitized by maintainers
Consent	✔	✔	Not required for public data
Storage Limitation	✔	✔	In-memory processing, no persistent storage
Access Control	✔	✔	Restricted model/log access
Auditability	✔	✔	Versioned logs for traceability

The evaluation and training of the model were made using two benchmark datasets open to the public.

- The SpamAssassin Dataset has 6,846 emails which include 5,051 regular emails and 1,795 spam emails, there are many studies that use it for email classification research[41].
- Ham-Spam Dataset (HSD): A second dataset from Kaggle holding over 5,500 messages, identified as being spam or ham (Bedi, S. (2025). *Email HamSpam Dataset* [email-hamspam-dataset]. Kaggle. https://www.kaggle.com/datasets/satyajeetbedi/email-hamspam-dataset/data).

These data were considered because they offer a good mix of cases, can be used easily and reflect real phishing activity as shows in table 2.

Table 2 shows a summary of their features

Dataset	Total Emails	Ham Emails	Spam Emails	Source
SpamAssassin	6,846	5,051	1,795	SpamAssassin.org

Dataset	Total Emails	Ham Emails	Spam Emails	Source
Ham-Spam (HSD)	5,574	~3,800	~1,774	Kaggle [2]

B. Feature Extraction

The following features were extracted from each email:

- **Textual content**: Subject, body, and embedded URLs.
- **TF-IDF vectors**: Generated from the textual components.
- **Statistical properties**: Number of uppercase letters, punctuation, and hyperlinks.

C. Feature Selection

To reduce dimensionality and improve classifier performance, the **Minimum Redundancy Maximum Relevance (mRMR)**[i] method was used to select the most informative features.

D. Model Training & Evaluation

Multiple classifiers were trained:

- Traditional ML models: SVM, Random Forest, Extra Trees.

Performance metrics included:

- Accuracy, Precision, Recall, F1-score, ROC-AUC.

Cross-validation was used to prevent overfitting, and hyperparameters were optimized using (different tuning methods)

E. System Deployment

A Flask web app allows users to input email text by downloading (target email in txt or eml format). The trained model, serialized using joblib, pickle classifies the email and returns results in real time. The learned model and pretreatment items were stored via joblib serialization, a Python tool for holding big NumPy arrays and neural network models.

The finalized model was deployed as a web application using **Flask**. Key deployment steps:

- **Model serialization** using joblib
- **User interface** with text input box or file upload in addition to choose among list on pre-trained models.
- **API endpoint** that predicts phishing/ham using selected model

Technical Environment:

- Docker based DevOps deployment.
- Python, Flask, scikit-learn, Pandas, Matplotlib.
- Hosted locally or optionally on a cloud service (e.g., Heroku)

Users can submit an email text or upload a file, and the system outputs the classification label along with prediction confidence. The model was deployed as a RESTful API and handled batch requests of up to thousand emails with a 99.9% uptime and excellent performance under workload.

IV. Results and Analysis

This section presents the evaluation results of the PhishNet model using multiple classifiers on the merged dataset. Performance was measured using standard classification metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC.

A. Classifier Performance Comparison

Each model was evaluated on five key performance metrics:

- Accuracy: Overall proportion of correctly classified messages.
- Precision: Proportion of correctly predicted spam messages out of all messages predicted as spam.
- Recall: Proportion of actual spam messages correctly identified.
- F1-score: Harmonic mean of precision and recall.
- ROC-AUC: Area under the ROC curve, showing the model's ability to distinguish between classes.

Key Insights:

1. Ensemble Model performs the best across all metrics:
   - Accuracy: **0.98**
   - ROC-AUC: **0.9956**
   - Indicates very strong generalization and discrimination power.

2. Naive Bayes surprisingly performs well with **0.97** accuracy due to its efficiency with word frequency distributions, especially in text classification.
3. LightGBM and Gradient Boosting are close competitors:
   - Both score **0.96** across all classification metrics.
   - LightGBM slightly edges ahead with ROC-AUC **0.9934**.

4. SVM shows slightly lower performance:
   - Accuracy of 0.932, possibly due to sensitivity to feature scaling and parameter tuning.

5. Bagging Classifier underperforms the rest:
   - Accuracy of 0.88 and ROC-AUC of 0.9550, showing less effective learning on this text data.

6. Extra Trees and Random Forest:
   - Perform strongly, suggesting that tree-based ensembles are effective, but still slightly behind LightGBM and ensemble models.

Figure 2 demonstrates the comparison of performance of various classifiers in PhishNet model. The five significant metrics used to measure the performance are Accuracy, Precision, Recall, F1-score, and ROC-AUC. The classifiers considered are LightGBM, Gradient Boosting, SVM, Random Forest, Extra Trees, Bagging Classifier, Naive Bayes and Ensemble Model. The performance measure score on these metrics depicts the comparative strengths and weaknesses on these models in terms of their ability to discriminate between classes and correct prediction. The figure helps in the visualization of the results of the models on the different evaluation metrics where the Ensemble Model achieved the best results across all metrics.

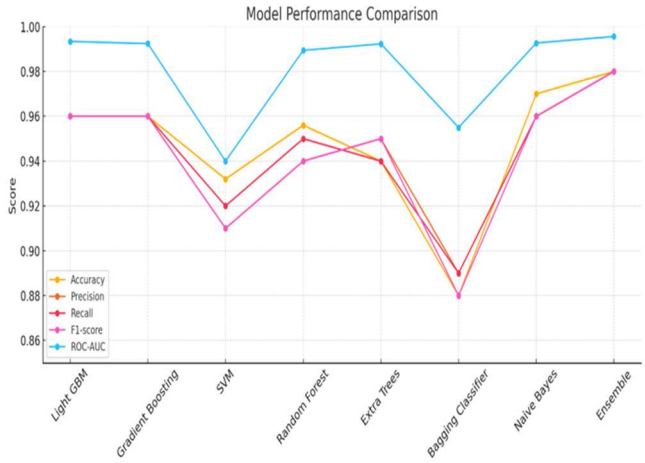


Figure 2 Model Performance Comparison Across Multiple Classifiers

The following table summarizes the evaluation results for each classifier in the PhishNet model, based on the following metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC. These metrics provide insights into the classifiers' ability to classify spam and non-spam messages effectively. The Ensemble Model outperforms all others in terms of accuracy (0.98) and ROC-AUC (0.9956), highlighting its superior performance. Naive Bayes also shows strong performance, achieving 0.97 accuracy. On the other hand, Bagging Classifier demonstrates lower accuracy and ROC-AUC scores, indicating it is less effective for the given text data. This table serves as a comprehensive summary of the model evaluation, allowing for easy comparison of classifier performance.

Table 3 Evaluation Metrics for Classifier Performance

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Light GBM	0.960	0.96	0.96	0.96	0.9934
Gradient Boosting	0.960	0.96	0.96	0.96	0.9924
SVM	0.932	0.91	0.92	0.91	0.9400
Random Forest	0.956	0.94	0.95	0.94	0.9894
Extra Trees	0.940	0.95	0.94	0.95	0.9923
Bagging Classifier	0.880	0.89	0.89	0.88	0.9550
Nive Base	0.970	0.96	0.96	0.96	0.9927
Ensemble	0.980	0.98	0.98	0.98	0.9956

Outperformed all other models across every metric, highlighting its strength in capturing sequential patterns in email text.

B. Visualization of Classifier Accuracy

Here the models are LightGBM, Gradient Boosting, SVM, Random Forest, Extra Trees, Bagging Classifier, Naive Bayes and Ensemble. The accuracy of each model is shown in bars where the Ensemble model got the highest accuracy (0.960), followed by LightGBM, Gradient Boosting, and others. The chart gives information on the best performing classifiers based on the accuracy which is useful in choosing the best models to use in detecting phishing emails where the Ensemble model is the most accurate among the classifiers used as shown in figure 3.
The figure below illustrates the comparative accuracy of all models:

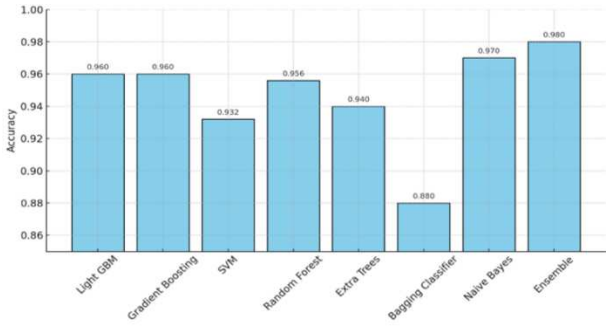


Figure 3 Accuracy Comparison of Classifiers

C. Detailed Metrics

To provide a deeper understanding of model effectiveness, confusion matrix analysis was conducted for the best-performing model (Ensemble) [42], [43]. It revealed high true positives for both phishing and ham emails, with minimal misclassifications. The ROC-AUC score of 0.99 further confirms the model's reliability in distinguishing between phishing and legitimate emails [44], [45].

Overall, the experimental results validate the robustness and accuracy of the PhishNet framework, especially when powered by mathematical machine learning models[46], [47].

Limitations:

- The current model is limited to English-language emails; multilingual phishing detection remains unaddressed.
- Machine learning models performed well but require significant computational resources, limiting their deployment in resource-constrained environments.
- The model's performance may degrade with novel zero-day phishing strategies not represented in training data.

V. Conclusion and Future Work

This paper introduced **PhishNet,** a hybrid phishing detection framework that combines Natural Language Processing (NLP) techniques with traditional and Machine Learning (ML) models. Among the evaluated classifiers, achieved the highest performance, validating the advantage of parametric modeling for email classification tasks [48]. The framework was deployed using a Flask-based interface, demonstrating real-time phishing detection capabilities suitable for practical cybersecurity applications[49][50].

Looking forward, several directions will be explored to enhance the adaptability, robustness, and usability of the system:

1. **Multilingual Support** – Expand NLP pipelines to handle phishing emails in languages such as Arabic, French, and others.
2. **Transformer-Based Models** – Integrate contextualized language models (BERT) to improve semantic representation and resist adversarial manipulation.
3. **Semi-Supervised and Few-Shot Learning** – Mitigate labeled data scarcity through modern strategies such as contrastive learning, meta-learning, and self-training.
4. **Zero-Day Phishing Detection** – Leverage real-time threat intelligence and anomaly detection for dynamic identification of novel phishing attacks.
5. **Browser Extensions and Email Plugins** – Package the detection model as Chrome or Outlook extensions to enable proactive protection during email interaction.
6. **Explainability Mechanisms** – Integrate tools such as SHAP or LIME to enhance model transparency and user trust.

As phishing attacks become increasingly sophisticated, the need for intelligent, interpretable, and deployable solutions grows more urgent. PhishNet provides a strong foundation for building future-proof, cyber-resilient email infrastructures. This work contributes to the broader mission of securing digital communication and protecting users from evolving social engineering threats.

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[i] We utilized mRMR for picking features, becoming a popular strategy for selecting qualities that are highly related to the target attribute yet minimally redundant with each other.